

## SCHOOL OF ENGINEERING

Diploma in Electronics & Computer Engineering (EGDF20)

### Wireless Communication & Networking (EGE321)

#### **Tutorial 3: Communication System Architecture**

- 1) Name two types of error that commonly occur in digital communication systems and state the underlying causes of the errors.

Ans:

The two types of error are:

Random Error – due to thermal noise.

Burst Error – due to multipath fading.

- 2) Give two reasons why baseband filtering is required in digital communications.

Ans:

To constraint signal bandwidth to improve bandwidth efficiency

To filter off unwanted sidelobes to prevent interference to adjacent channels.

- 3) What is the function of encryption?

Ans:

**Encryption** is the process of translating plain text data (plaintext) into something that appears to be random and meaningless (ciphertext).

privacy : to prevent unauthorized persons from extracting information from the channel (eavesdropping)

authentication : to prevent unauthorized persons from injecting information into the channel (spoofing)

- 4) The transmitter system of digital communication system can support a data rate of 16kbps within a bandwidth of 20 kHz by encoding 2 bits in each symbol.

- (i) What is the bandwidth efficiency of the radio link?

$$\text{BW efficiency} = \text{data rate} / \text{bandwidth} = 16\text{k}/20\text{k} = 0.8 \text{ bps/Hz}$$

- (ii) What is the symbol rate?

$$\text{Symbol rate} = \text{Data rate} / \log_2 M = 16\text{k}/2 = 8\text{ksym/s}$$

- 5) State Nyquist sampling theorem.

Ans:

Nyquist Sampling Theorem: The sampling frequency need to be  $> 2$  times the maximum analogue signal at the input of the ADC to avoid aliasing.

- 6) What is the function of the channel encoder in a digital transmitter?

Ans:

Channel encoder: to combat channel impairments by adding redundancy bits for error detection and correction.

- 7) A mobile radio can support a data rate of 16kbps within a bandwidth of 25 kHz by encoding 2 bits in each symbol. What is the bandwidth efficiency of the radio link? What is the symbol rate?

BW efficiency = data rate / bandwidth =  $16\text{k}/25\text{k} = 0.64$  bps/Hz

Symbol rate = Data rate /  $\log_2 M = 16\text{k}/2 = 8\text{ksym/s}$